

Project Report

Project Title : Global Superstore Sales Performance Dashboard

Tool Used : Tableau

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Date of Submission : March 2026

2. Introduction / Project Overview

This project involves building an interactive dashboard using Tableau to analyze the **Global Superstore** sales dataset. The dataset represents sales transactions from a fictitious global retail superstore operating across multiple regions and markets from 2011 to 2014 (or similar period based on data).

Background: Retail businesses generate large volumes of sales, profit, customer, and shipping data. Understanding patterns in this data helps identify high-performing areas, underperforming regions/products, and opportunities for improvement.

Purpose of Analysis: To create a visual, interactive dashboard that allows stakeholders (managers, executives) to quickly explore sales performance, profitability, regional contributions, product performance, customer segments, and yearly trends.

Business Problem: The superstore faces challenges in identifying which regions/products/segments drive profit vs. losses, understanding sales growth patterns, and optimizing decisions like inventory, marketing, and discounts.

The dashboard addresses these by providing key insights at a glance.

3. Objectives of the Project

- Analyze sales and profit performance by region/market to identify the highest contributing area.
- Identify the top 10 products by revenue and understand their profitability.
- Evaluate profit margin across customer segments (Consumer, Corporate, Home Office) to find the most efficient segment.
- Visualize yearly sales and profit growth trends to assess business progress over time.
- Provide actionable business recommendations based on data-driven findings.

4. Dataset Description

Dataset Name : Global Superstore (or Global_Superstore2)

Source : Kaggle (public dataset, commonly used for Tableau practice and analysis)

Number of Rows & Columns : Approximately 51,000 rows (order line items) and 24 columns (varies slightly by version; standard Orders sheet used).

Key Data Fields Description:

Column Name	Description
Order Date	Date when the order was placed
Ship Date	Date when the order was shipped
Ship Mode	Shipping method (Standard, Second Class, etc.)
Customer ID / Name	Unique customer identifier and name
Segment	Customer type (Consumer, Corporate, Home Office)
Country / City	Location of the customer
Region / Market	Geographic region or market (e.g., APAC, EU, USCA, Canada)

Column Name	Description
Category	Product category (Technology, Furniture, Office Supplies)
Sub-Category	More detailed product grouping
Product Name	Specific product name
Sales	Revenue generated from the sale
Profit	Profit after costs and discounts
Quantity	Number of units sold
Discount	Discount percentage applied

Main sheet used: Orders (line-level data).

5. Data Cleaning / Data Preparation

- Connected to the Excel/CSV file in Tableau.
- Checked for null values – minor nulls in postal code ignored (not used in analysis).
- Ensured date fields (Order Date, Ship Date) recognized as Date type.
- Created calculated fields:
- **Profit Margin %** = $\text{SUM}([\text{Profit}]) / \text{SUM}([\text{Sales}])$ (formatted as percentage).
- **Avg Shipping Days** = $\text{AVG}(\text{DATEDIFF}('day', [\text{Order Date}], [\text{Ship Date}]))$.
- No major duplicates removed (dataset mostly clean).
- Year extracted from Order Date for time-based analysis.
- No advanced ETL needed as Tableau handled most transformations.

6. Data Model

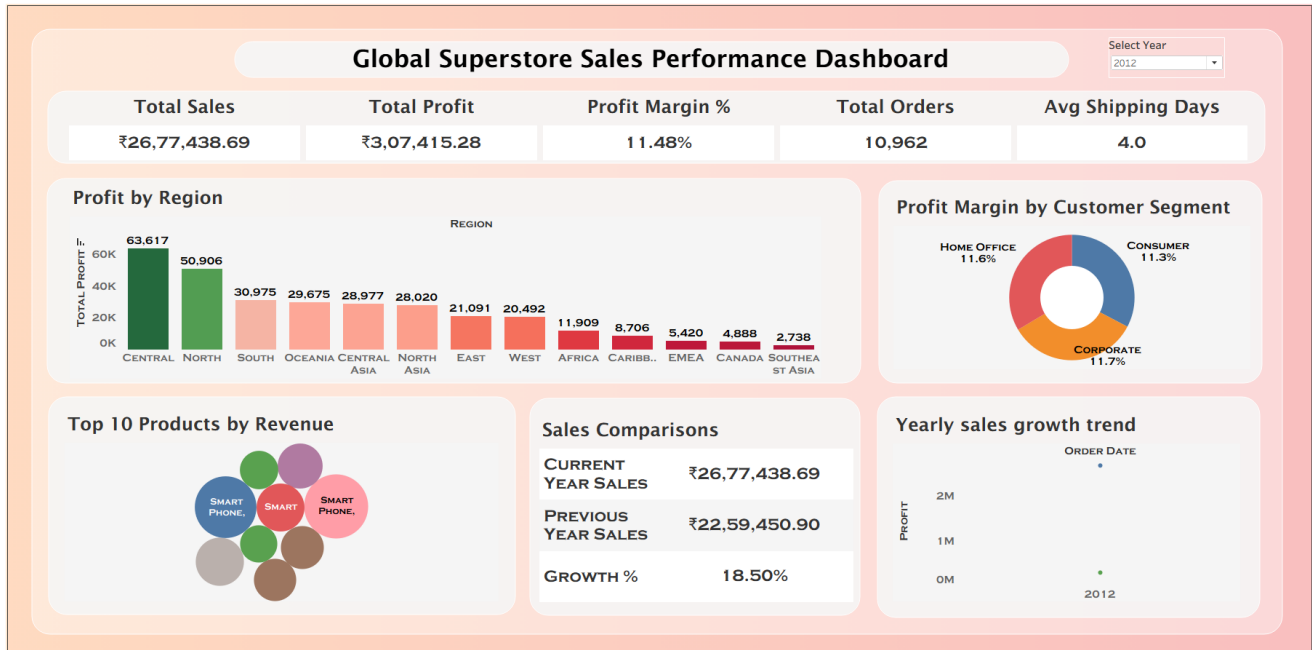
Single table dataset (Orders sheet), no multiple tables or joins required. All analysis done on one main data source.

7. Calculated Fields (Tableau)

- **Profit Margin %**
 $\text{SUM}([\text{Profit}]) / \text{SUM}([\text{Sales}])$
 (Formatted as Percentage)

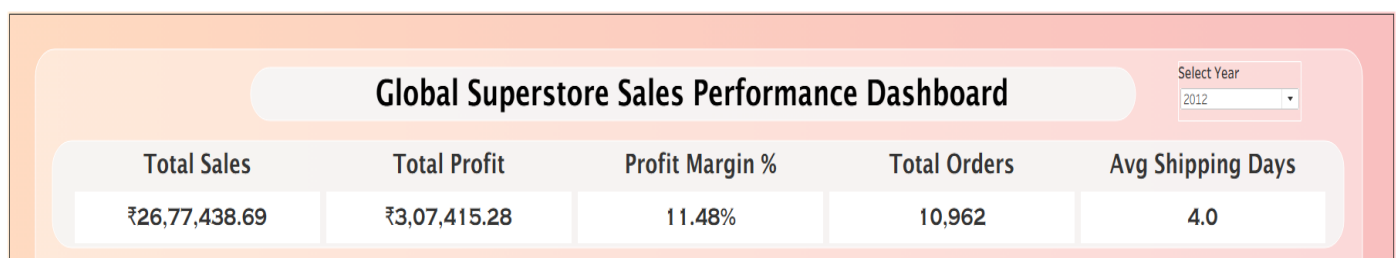
- **Avg Shipping Days**
 $\text{AVG}(\text{DATEDIFF}(\text{'day'}, [\text{Order Date}], [\text{Ship Date}])))$

8. Dashboard Design

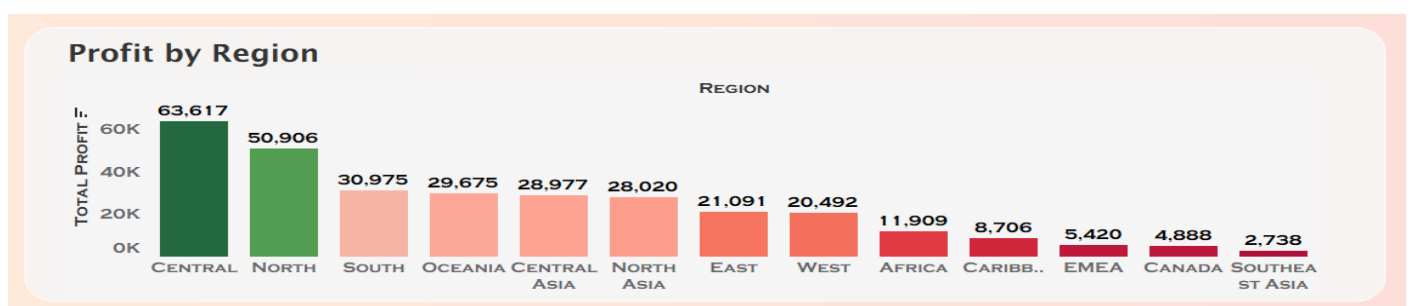


The dashboard is built with 4–5 main sheets + KPI cards:

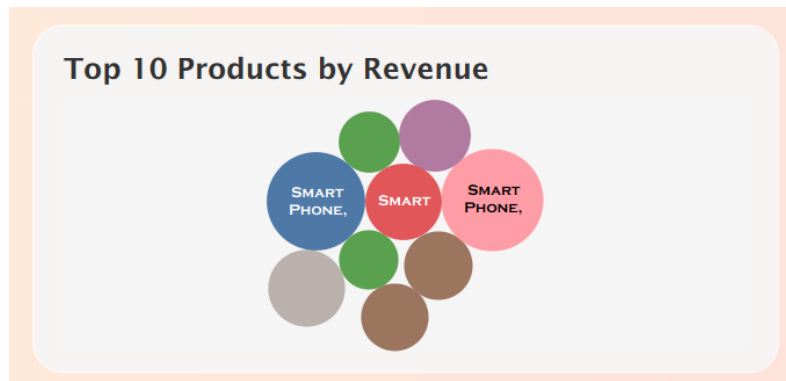
- **KPI Cards (Top):** Total Sales, Total Profit, Profit Margin %, Avg Shipping Days, Total Orders.



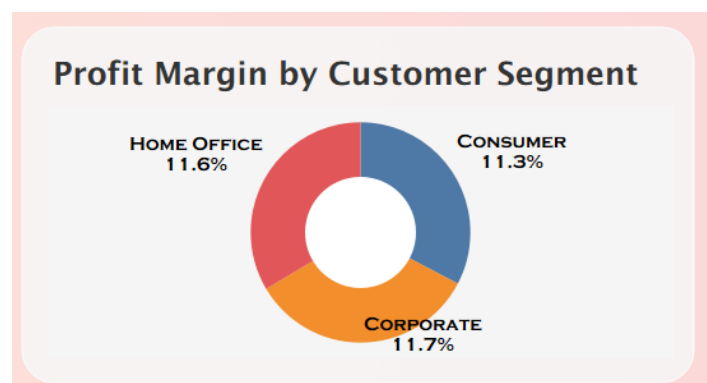
- **Profit by Region/Market** — Bar chart (or filled map attempt) to show highest profit area.



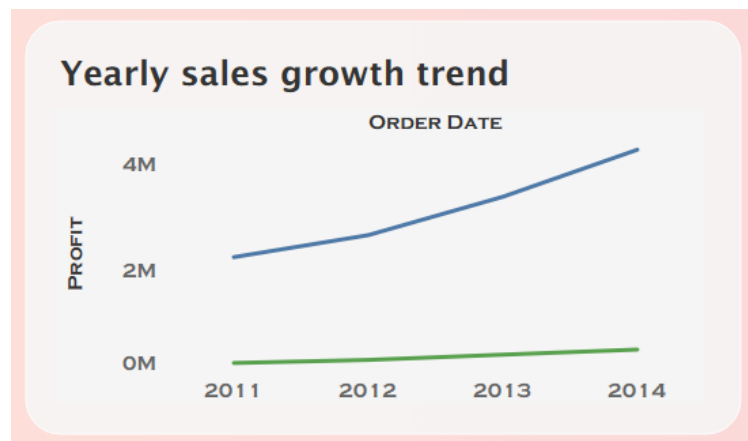
- **Top 10 Products by Revenue** — Packed Bubbles chart (size = sales, color = category/profit).



- **Profit Margin by Customer Segment** — Donut/Pie chart (size = margin %, color = segment).



- **Yearly Sales Growth Trend** — Dual-axis line chart (Sales & Profit lines + trend line, optional YoY % line).



Interactivity & Global Filtering



The dashboard includes a **Year Filter** (parameter-based dropdown) that allows users to dynamically select a specific year (2011, 2012, 2013, 2014 or All Years). This filter is applied globally, meaning when a year is selected, **all visuals on the dashboard update automatically**:

- KPI cards (Total Sales, Total Profit, Profit Margin %, Avg Shipping Days, Total Orders) reflect data only for the selected year
- Profit by Region bar chart shows profit distribution for the chosen year
- Top 10 Products by Revenue (Packed Bubbles) displays the top revenue products for that year
- Profit Margin by Customer Segment (Donut/Pie chart) updates segment-wise profit margins for the selected period
- Yearly Sales Growth Trend line chart highlights or filters data to the chosen year

Implementation Details:

- Created using a quick filter on the **Order Date** field (Year level)
- Applied to all worksheets via **Apply to Worksheets** → **All Using This Data Source**
- Filter style: Dropdown (single-select) with "All" option enabled
- Placement: **Top-right corner** of the dashboard (prominent, easy to access, does not obstruct main visuals)

Sales Comparisons – Year-over-Year Overview

Sales Comparisons	
CURRENT YEAR SALES	₹26,77,438.69
PREVIOUS YEAR SALES	₹22,59,450.90
GROWTH %	18.50%

One of the key interactive elements in the dashboard is the **Sales Comparison card**, which dynamically shows the current year sales, previous year sales, and the year-over-year (YoY) growth percentage.

This KPI card helps stakeholders quickly understand business growth at a glance.

Key Metrics Displayed:

- **Current Year Sales:** ₹26,77,438.69

- **Previous Year Sales:** ₹22,59,450.90
- **Growth %:** 18.50%

The card visually highlights positive growth with green coloring (or upward arrow icon if implemented) to emphasize improvement in sales performance compared to the previous year.

Calculation Logic (behind the scenes):

- Current Year Sales = SUM(Sales) for the selected year (via Year filter/parameter)
- Previous Year Sales = SUM(Sales) for the previous year (using table calculation or parameter offset)
- Growth % = ((Current Year Sales - Previous Year Sales) / Previous Year Sales) × 100

Visualizations Used:

Visualization	Purpose
KPI Cards	Quick overview of key metrics
Bar Chart	Compare profit by region
Packed Bubbles	Show top 10 products by revenue (size-based)
Donut/Pie Chart	Distribution & comparison of profit margin by segment
Line Chart	Show sales/profit trend over years

Dashboard size fixed to ~1200 × 800 px for presentation fit on laptop screens. Global filters (Year, Region, Segment) applied for interactivity.

9. Key Insights / Findings

- **Highest Profit Region/Market:** APAC generates the highest profit (leads in sales volume and margin from Technology products), while Canada ranks lowest.
- **Top 10 Products by Revenue:** High-ticket items like Canon copiers, Cisco/Apple/Samsung phones, and HON/SAFCO chairs dominate (Technology category drives most revenue).
- **Customer Segment Profit Margin:** Corporate or Home Office segment has the highest profit margin % (efficient buying, fewer discounts), while Consumer has highest total profit (high volume but lower margin).

- **Yearly Sales Growth Trend:** Clear upward trend from 2011 to 2014; sales and profit steadily increased (Technology category main driver; positive YoY growth most years).
- **Other observations:** Some high-revenue products show losses due to discounts; shipping delays impact customer satisfaction in certain modes.

10. Business Recommendations

- Prioritize marketing and inventory in **APAC** market (highest profit contributor).
- Focus sales efforts on **Technology** category and top revenue products (e.g., copiers & smartphones).
- Reduce discounts in low-margin categories (Furniture sometimes loss-making) to improve overall profitability.
- Target **Corporate** and **Home Office** segments more aggressively for higher margins.
- Investigate shipping modes to reduce average days and improve customer satisfaction.
- Monitor yearly trends closely – continue growth strategies that drove 2011–2014 increase.

11. Challenges Faced

- Initial issues with geographic mapping (Region/Market fields not fully recognized by Tableau for filled maps – switched to bar charts).
- Overlapping labels in packed bubbles – adjusted size/font for readability.
- Dual-axis synchronization for trend charts (sometimes axis not aligning – fixed by manual sync).
- Dataset variations across versions (Global_Superstore2 had slightly different region/market groupings).

12. Conclusion

This Tableau dashboard successfully visualizes and analyzes the Global Superstore sales data, highlighting key performance areas like regional profitability, product revenue leaders, segment margins, and growth trends. The interactive design enables quick insights for data-driven decisions. The project demonstrates effective use of Tableau visualizations, calculated fields, and dashboard design principles.

13. Future Improvements

- Integrate real-time or updated data feeds.
- Add forecasting (using Tableau's built-in forecast feature).
- Include predictive analytics (e.g., customer churn or demand prediction).
- Add more filters (e.g., Sub-Category, Ship Mode).
- Export to Tableau Server/Public for sharing.

14. References

- Dataset: Global Superstore on Kaggle (various versions, e.g., by Vivek468 or similar).
- Tableau Official Documentation & Help (tableau.com).
- Tableau Public sample dashboards (Super Sample Superstore by Ryan Sleeper).
- Kaggle notebooks & community analyses for inspiration.